PRACTICE SHEET

1. A vertical pole with height more than 100 consists of two parts, the lower being one – third of the whole. At a point on a horizontal plane through the foot and 40m from it, the upper part subtends an angle whose tangent is1/2. What is the height of the pole?

(a) 110m (b) 200m
(c) 120 m (d) 150m

2. The angle of elevation of the top of a pillar of height h at a point on the ground at a distance x from the pillar is 30°. On walking a distance 'd' towards the pillar the angle of elevation becomes 60°. Then, which one of the following is correct?

(a) $x = d + h$ (b) $x = \frac{3d}{2}$

(c) $x = \frac{5d}{4}$ (d) $x = 2d$

3. The angle of elevation of the top of a tower EF (F being the foot of the tower) as seen from a point A which is on the same level as F, is α. On advancing towards the foot of the tower the angle of elevation of the top of the tower as seen from a point B such that AB = x, is β. If BF = y, h is the height of the tower and $\alpha + \beta = \pi/2$, then which one of the following is correct?

(a) $h^2 = x^2 + xy$ (b) $h = y^2 + xy^2$
(c) h2 = y2 + xy (d) h = y + x2y

8. A man observes the elevation of a balloon to be 30°. He then walks 1 km towards the balloon and finds that the elevation is 60°. What is the height of the balloon?

(a) 1/2 km (b) $\sqrt{3}$ /2 km
(c) 1/3 km (d) 1 km

9. The angle of elevation from a point on the bank of a river of the top of a temple on the other bank is 45°. Retreating 50m, the observer finds the new angle of elevation as 30°. What is the width of the river?

(a) 50m (b) $50\sqrt{3}$m
(c) $50/(\sqrt{3}-1)$m (d) 100m

10. Looking from the top of a 20 m high building, the angle of elevation of the top of a tower is 60° and the angle of depression of its bottom is 30°. What is he height of the tower?

(a) 50m (b) 60m
(c) 70m (d) 80m

11. Two houses are collinear with the base of a tower and are at distance 3m and 12m from the base of the tower. The angles of elevation from these two houses of the top of the tower are complementary. What is the height of the tower?

(a) 4m (b) 6m
(c) 7.5m (d) 36m

12. The angle of depression of vertex of a regular hexagon